

A Counterexample to the Reverse Exact-Dominant Narrowness Question

Automatic Math Discovery Smoke Test

July 1, 2026

Source and Extracted Question

The source is Nariman Abasov, Abd El Monem Megahed, and Marat Pliev, *Dominated operators from a lattice-normed space to a sequence vector lattice*, arXiv:1508.03275. In Section 5 the authors prove that, under their Banach–Kantorovich hypotheses, order narrowness of a dominated linear operator T implies order narrowness of its exact dominant $|T|$. The following remark asks whether the converse is true:

Does the order narrowness of the operator $|T|$ imply the order narrowness of T ?

have $|T|(e) = |T|(e_+) - |T|(e_-)$. Thus, if $e_+ = f_1^\alpha \sqcup f_2^\alpha$ and $e_- = g_1^\alpha \sqcup g_2^\alpha$ are necessary nets of decompositions, then

$$\begin{aligned} & \left| |T|(f_1^\alpha + g_1^\alpha) - |T|(f_2^\alpha + g_2^\alpha) \right| = \\ & \left| |T|f_1^\alpha - |T|f_2^\alpha + |T|g_1^\alpha - |T|g_2^\alpha \right| \leq \\ & \left(\left| |T|(f_1^\alpha - f_2^\alpha) \right| + \left| |T|(g_1^\alpha - g_2^\alpha) \right| \right) \xrightarrow{(o)} 0 \end{aligned}$$

and $e = (f_1^\alpha + g_1^\alpha) \sqcup (f_2^\alpha + g_2^\alpha)$ is a desirable net of decompositions. □

Remark 12. *This is an open question. Does the order narrowness of the operator $|T|$ implies the order narrowness of the T ? A particular case was proved in [17] Theorem 5.1.)*

Counterexample

Let the measure space be the Lebesgue interval $([0, 1], \mathcal{L}, m)$. Put

$$E = L_\infty[0, 1], \quad V = L_\infty[0, 1], \quad p(f) = |f|.$$

Thus V is the usual vector lattice $L_\infty[0, 1]$, regarded as a lattice-normed space over itself. It is decomposable, *bo*-complete, Dedekind complete, atomless, and has the strong Freudenthal property.

Let

$$F = \mathbb{R}, \quad W = L_1[0, 1], \quad q(g) = \|g\|_1.$$

Every Banach space is a Banach–Kantorovich space over $\mathbb{R}$ with its norm as the vector norm. Define

$$T : V \longrightarrow W, \quad Tf = f.$$

This is the natural inclusion $L_\infty[0, 1] \subset L_1[0, 1]$.

Theorem 1. *The operator $T : L_\infty[0, 1] \rightarrow L_1[0, 1]$ is a dominated linear operator whose exact dominant is order narrow, but T is not order narrow.*

Proof. For $e \in E_+$, define

$$S(e) = \int_0^1 e(t) dt.$$

Then for every $f \in V$,

$$q(Tf) = \|f\|_1 = \int_0^1 |f(t)| dt = S(p(f)),$$

so S is a dominant of T .

It is the exact dominant. Indeed, if $R : E \rightarrow \mathbb{R}$ is any positive dominant, then for every $e \in E_+$ we may take $f = e$ and obtain

$$S(e) = \|e\|_1 = q(Te) \leq R(p(e)) = R(e).$$

Thus $S \leq R$, proving that $S = |T|$.

Next S is order narrow. Let $e \in E$. Write $e = e^+ - e^-$. Since the Lebesgue measure algebra is atomless, there are measurable sets $A_+ \subset \{e > 0\}$ and $A_- \subset \{e < 0\}$ such that

$$\int_{A_+} e^+ dm = \frac{1}{2} \int e^+ dm, \quad \int_{A_-} e^- dm = \frac{1}{2} \int e^- dm.$$

Put $A = A_+ \cup A_-$. Then

$$e = e\chi_A \sqcup e\chi_{A^c}$$

is a mutually complemented fragment decomposition in E , and

$$S(e\chi_A) - S(e\chi_{A^c}) = 0.$$

So the constant net given by this decomposition witnesses order narrowness of S at e .

Finally T is not order narrow. Consider $v = \mathbf{1} \in V$. If

$$\mathbf{1} = v_1 \sqcup v_2$$

is a mutually complemented fragment decomposition in V , then necessarily

$$v_1 = \chi_A, \quad v_2 = \chi_{A^c}$$

for some measurable $A \subset [0, 1]$. Hence

$$q(Tv_1 - Tv_2) = \|\chi_A - \chi_{A^c}\|_1 = 1.$$

Thus no net of mutually complemented fragment decompositions of $\mathbf{1}$ can make $q(Tv_1 - Tv_2)$ converge to 0. Therefore T is not order narrow. $\square$

Why This Answers the Source Question

The construction lies inside the structural setting of the theorem preceding the source remark:

$$(V, E) = (L_\infty, L_\infty), \quad (W, F) = (L_1, \mathbb{R}),$$

with E atomless and Dedekind complete and F Dedekind complete. The exact dominant exists explicitly and is order narrow, yet the original dominated linear operator fails order narrowness. Hence the reverse implication asked in the remark is false.

Novelty Notes

The local run indexes were searched for the arXiv id 1508.03275, the paper title, and phrases around **exact dominant** and **order narrow**. No duplicate solution packet was found. Local source search found related vector-lattice and abstract-Uryson modulus results, but those do not remove this lattice-normed-space counterexample.

Bounded web searches on July 1, 2026 for the exact question phrase, the paper title, and close **order narrow exact dominant lattice-normed** phrases found the source arXiv page and a related orthogonally additive paper, but no exact published resolution of the reverse implication.

References

- [1] N. Abasov, A. E. M. Megahed, and M. Pliev, *Dominated operators from a lattice-normed space to a sequence vector lattice*, arXiv:1508.03275.